

Week 4 Notes

STAT 251 SECTION 01

The normal distribution and z -scores

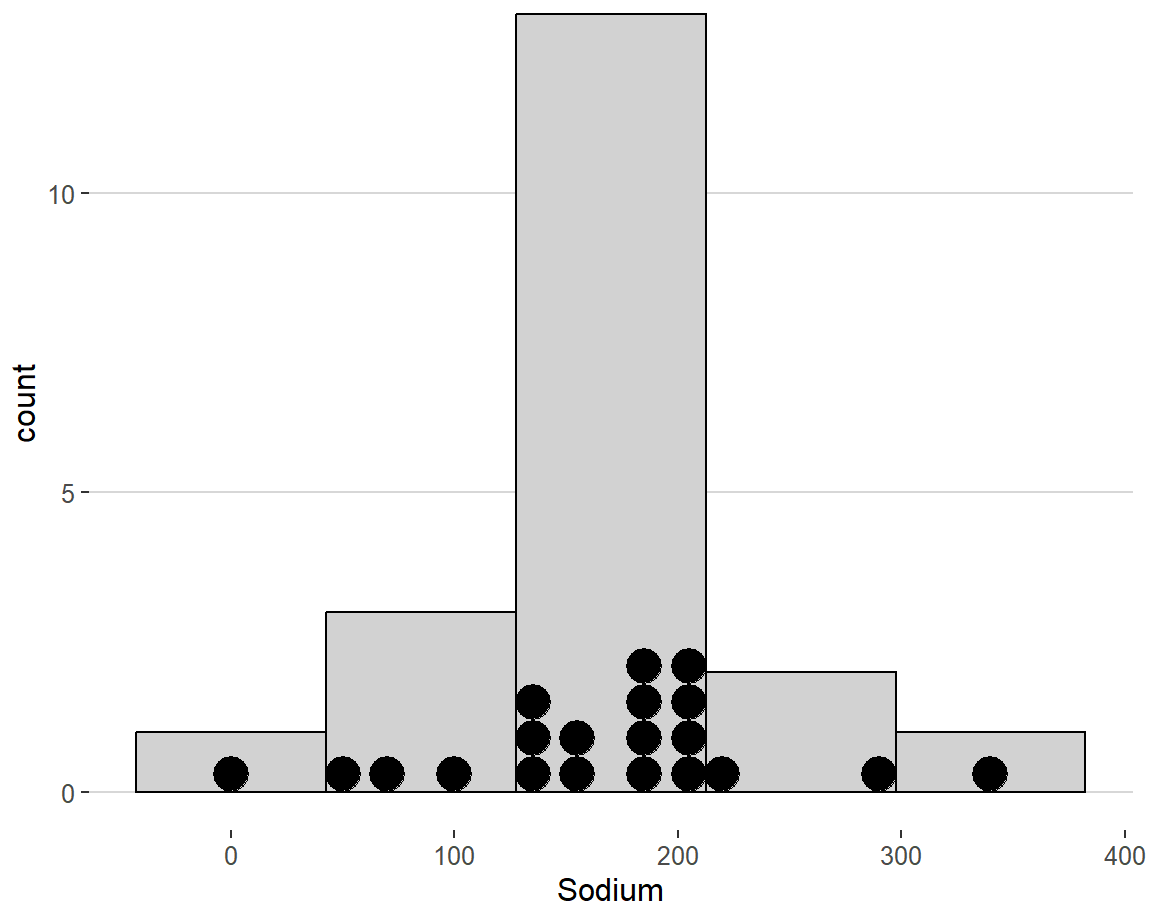
Week 3 finished our tour of **descriptive statistics**: we can now describe the shape of a distribution, its center, its spread, and the position of any individual value within it. Everything on Exam 1 comes from that material.

This week starts a different question. So far, “far from the center” has been measured in the units of whatever we happened to be measuring — points on an exam, grams of sugar, milligrams of sodium. That makes it impossible to compare across variables: is a student 12 points above the class mean further from typical than a cereal 90 mg above the sodium mean? The two numbers are not on the same scale, so the comparison is meaningless as stated.

The way out is to stop measuring distance in the original units and start measuring it in **standard deviations**. That single idea gives us z -scores, and it is what makes the normal distribution as useful as it is.

Measuring a deviation in standard deviations

Recall the twenty US breakfast cereals from the Week 3 notes. Below is a dot plot of their sodium content overlaid on a histogram of the same variable.



- What is the largest positive deviation in this dataset? How does this deviation compare to the standard deviation of the entire dataset?
- The largest deviation belongs to the cereal Raisin Bran which has a sodium content of 340 mg. Its deviation is $340 - 167 = 173$. The standard deviation for the whole dataset is about 77mg, so this deviation is about $173/77 \approx 2.25$ standard deviations above the mean. This indicates that Raisin Bran has a much higher sodium content than majority of the cereal brands in this dataset.

That last sentence is doing something new, and it is worth naming. We did not say Raisin Bran was 173 mg above the mean and leave it there — we divided that deviation by the standard deviation and reported the answer as “about 2.25 standard deviations above the mean”. Dividing a deviation by s strips the units off it, and what is left is a number that means the same thing no matter what was being measured. It is the quantity we will call a **z -score**:

$$z = \frac{x - \bar{x}}{s}$$

A cereal at $z = 2.25$ and a student at $z = 2.25$ are equally unusual within their own distributions, even though one is measured in milligrams and the other in exam points.

